



Simplified thermal model of the ITER magnet system



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ABSTRACT

A simplified thermal model of the ITER magnet system has been developed to capture the essence of the magnet heat load dynamics without the need for extensive computations. Idealization of the magnets has been made using mainly two standard types of elements, solids and tubes. No Navier–Stokes equations have been solved for the hydraulics, but instead a simple transport model with approximation for pressure evolution has been used. The model was implemented in C language and used to investigate the important features needed to implement a computationally efficient and fast magnet thermal model capturing overall behavior in terms of superconductor cooling channel description (thermal coupling with jackets, presence of the conductor, importance of the central channel, etc.). Furthermore, the model was benchmarked against validated simulation tools such as SuperMagnet and Vincenta using the ITER Central Solenoid normal operation scenario for comparison. Dynamics were shown to be reproduced in good agreement with results attainable with these more detailed codes, considering the high level of uncertainty on the input parameters, namely the heat transfer coefficients and the values of heat loads.

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1. Introduction

1.1. Context of this work

The fast changing magnetic fields required for the operation of the fusion reactor ITER (currently in design and construction) are achieved thanks to a magnet system based on superconducting coils. During the normal operation of the machine, the magnet structure and the conductor itself are subject to time-varying non-uniform thermal heat loads of diverse origins (nuclear radiation, eddy currents, Ohmic losses in joints, AC losses in superconductors, etc.). The ITER magnets are based on cable-in-conduit conductors (CICC); superconducting strands are disposed in bundles inside a pipe through which supercritical helium flows. The heat loads generated in the magnets are transported by the fluid to the cryoplant, composed of a set of heat exchangers that on the secondary side are in contact with a boiling helium bath. From there the heat is transferred to a refrigeration cycle that ultimately removes it from the circuit [9].

Thermal modeling of the ITER magnet system needs to account for two main aspects. Firstly, assessing conductor temperature

both along the entire length of the flow path, and over time during the transient, to verify that a sufficient thermal margin exists throughout the plasma scenario. Secondly, a good temporal (dynamic) estimate of the energy being carried away from the magnets to the cryoplant by the outlet helium flow is needed. In turn, this information can be used to assess the heat load the cryoplant needs to manage so that magnet inlet temperature is kept constant despite a heavily pulsating heat load. Models should be able to reproduce the magnets dynamics taking as an input the helium flow conditions provided by the cryoplant and a heat load scenario, and returning as an output the helium outlet conditions, that constitute the inlet of the cryoplant and its heat load condition (see Fig. 1).

The tools currently available to analyze the conductor margins are highly detailed models that describe thoroughly the system with most of its elements and their thermal interactions. They are capable of producing accurate distributions of temperature along the flow paths over time as required to assess the conductor operating margin. However, they tend to be extremely detailed (and hence slow) for the purpose of assessing the overall heat load and its effect on the magnet/cryoplant interface.

The intention behind this work is to develop reasonably accurate, yet very simplified, thermo-hydraulic models of the ITER magnets that could capture the overall transient heat load response as seen by the cryoplant, and to do so with as few elements as possible thus optimizing the computational efficiency. It is

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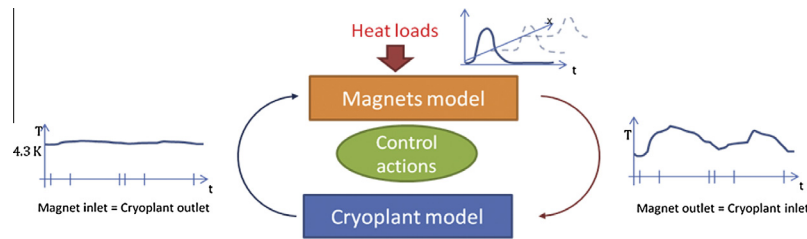


Fig. 1. Schematic of the magnet/cryoplant interface.

essential that these models represent as accurately as possible the time behavior transfer function from magnet heat load to cryoplant heat load, without much concern for calculating temperature distributions inside the magnet system itself. It is highly desirable that the execution be fast (preferably faster than real-time) so as to use the models in design iteration loops, control strategy exploration and evaluation and, eventually, as a predictive control tool in real time operation of the ITER machine.

1.2. Background on the ITER magnet system

In this section, the main design features of ITER magnets relevant to the development of the simplified thermal model are highlighted. A detailed description of the ITER magnets and their manufacturing process can be found in works by Mitchell et al. [6], Mitchell et al. [7] and Devred et al. [2].

In ITER's CICCs the superconducting strands are placed in bundles in the annular space, separated from the central channel by a stainless steel spiral. The central channel provides a low-resistance path to the helium flow, whereas fluid exchange is possible between the channel and the cable space. The conductor is contained in the stainless steel jacket, which can be either square or circular, depending on the magnets. As an example, a picture of the cross section of ITER Central Solenoid (CS) conductor is shown in Fig. 2.

Conductors are wound into flat pancakes, which are then stacked and connected to make coil modules. Coolant is injected by feeders every two pancakes in such a way that alternate pancakes have contrary helium flow directions. The feeders are placed at the innermost turn of the pancake. Helium is collected at the outermost turn.

To summarize, a magnet cross section is composed of parallel pipes separated by solid structures. Pipes on the same row have the same flow direction – either into or out of the cross section – but those in alternate rows are in counterflow.

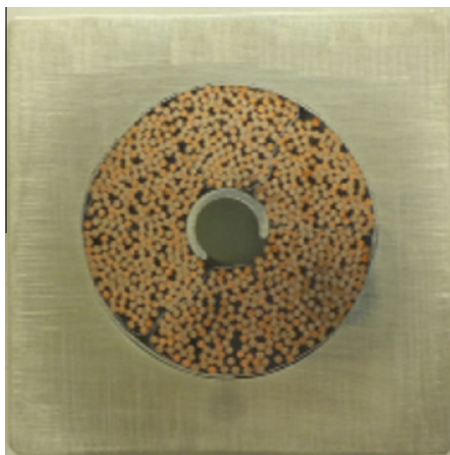


Fig. 2. Cross section of the ITER CS conductor.

Tubes in the magnet conductor are just a part of the cryogenic hydraulic network, which also includes the heat exchangers in the cryoplant, pumps, valves, distribution lines and feeders, connection volumes and other elements. These are, in turn, modeled using lumped parameter/flow models. One of the main objectives of this work is to create an equivalent model of the magnet (including its heat loads) for magnet/cryoplant interface integration studies.

2. Simplified thermal model

The model proposed in this work relies on a generic and simplified representation of a superconducting magnet system, its flow paths, heat capacities, and heat loads during operation. In a first stage an idealized simplified magnet cooling network needs to be created. The idealization may consist of combining several identical parallel branches into a single equivalent pipe, the unification of several structure parts, etc. The next step is to fit that idealization into the representation constructed with the elements presented in Section 2.1.

2.1. Elements of the model

The simplified model consists of an ideal hydraulic network representing the magnets and the cryogenic pipelines connecting them to the cryoplant. This network is constructed using three types of elements:

1. Tubes: elements that transport fluid, transferring inlet properties to the outlet with a transit time depending on mass flow rate, density and volume, incorporating heat input to the fluid and pressure changes.
2. Solids: structural elements that can accumulate thermal energy (heat capacity) and exchange it with tubes and other solids.
3. Connection nodes: elements of the hydraulic network with many inlets and/or outlets that mix incoming flow and provide a homogeneous output.

For purposes of modeling, each coil is subdivided into a number of elements. Each of these elements, called hereafter 'magnet element', has a particular cross section composed of solid and tube cross sections. Some of these tubes have parallel flow, so they can be considered equivalent and could be condensed into a single lumped element. Several solid parts may have a very similar temperature and heat load condition, so they could be lumped into a single solid element as well. Thus, a simplified cross section of the magnets can be represented using fewer tubes and solids than in the real magnet. Different magnet elements are then connected in series by interconnecting the outlets and inlets of their tubes. This way a simplified hydraulic network representing a magnet can be constructed.

Connection nodes and tubes are also used to construct the external hydraulic networks. Connection elements may or may

not have volume (the latter in the case ideal tube connections, bifurcations or junctions are necessary in the model simplification).

With the purpose of using different levels of simplification in the tube modeling, and also due to the presence of different heat load regimes on tubes, several types of tubes have been implemented:

- Conductor tube types:

- *Conductor tube 'c'*: this tube lodges a conductor, represented by a solid immersed within, and a central channel. There is thermal exchange (driven by temperature difference) between the annular space with these two immersed *sub-elements* and with the surrounding solid. Fig. 3 shows a schematic of this type of tube;
 - *Semi-simplified Conductor tube 's'*: in this tube there is no longer an immersed solid, and the heat load on the conductor is directly applied in the helium bulk of the annular space.
 - *Simplified Conductor tube 'z'*: this tube is a plain tube that can interact with a surrounding solid and also has a dynamic heat load applied directly on the fluid inside.
- Other tube types:
 - *Normal tube 'n'*: this tube exchanges heat only with the solid it goes through and proportionally to their temperature difference through a heat transfer coefficient that may depend on the mass flow rate in the tube;
 - *Ideal Heat Exchanger tube 'i'*: has an arbitrarily fixed wall temperature (that could vary in time) and the heat exchanged is proportional to the temperature difference between the wall and the fluid through a heat transfer coefficient that may depend on the mass flow rate in the tube;
 - *Heated tube 'h'*: this tube has an externally fixed heat source.

2.2. Governing equations

Each type of element has a particular behavior over time. The following sub-sections describe the mathematical model developed to capture the thermal dynamics of the system.

2.2.1. Solids

The solid elements are described by a simple lumped capacity model. Each solid is considered as a region with uniform temperature, which evolves according to the balance of energy:

$$M_i^{\text{solid}} c_i \frac{dT_i}{dt} = - \sum_j S_{ij}(T_i - T_j) - \sum_k E_{ik}(T_i - T_k) + F_i, \quad (1)$$

where m_i is the mass, C_i the specific heat and T_i the temperature of solid i . In this equation, the sum with index j represents the thermal exchange with other *solids* j in contact with it, and that one with in-

dex k , the heat flux exchanged by convection cooling in the *tubes* k that go through the *solid* i . Lastly, F_i is the total heat source in the volume of the solid i , which will be referred to as the heat load on the solid. S_{ij} and E_{ik} are the heat transfer coefficients.

2.2.2. Hydraulics and tubes

The hydrodynamics in this simplified model does not take into account momentum or energy balance equations. No Navier–Stokes equivalent equations are solved, but mass flow rate is an imposed external time-varying input of the model. Additionally, the simplifying assumption of incompressible flow with uniform fluid density along the hydraulic paths is introduced. The friction losses in the hydraulic network are neglected, and instead a uniform value of global pressure is calculated for each independent loop. Similarly to the model presented in the works by Rousset et al. [8], Hoa et al. [4] and Vallcorba et al. [10], pressure evolution is the result of the overall energy balance of the circuit in the form of heat transfer, $\dot{Q}_{\text{loop}}$ – the volume of the loop is held constant, which results in no external work:

$$\frac{du}{dt} = \frac{\dot{Q}_{\text{loop}}}{M_{\text{loop}}}, \quad (2)$$

where u is the average specific internal energy of the system, M_{loop} the total mass of helium in the closed loop and $\dot{Q}_{\text{loop}}$ is the net heat flow into the coolant along the closed loop circuit. Integration of this equation yields the values of u over time. A uniform pressure p in the circuit (neglecting hydraulic losses) can be calculated from the state equation of helium:

$$P = P(u, \rho). \quad (3)$$

The variations of this pressure are used to calculate pressure-induced changes in enthalpy. The helium properties implemented in the model are taken from the code Hepak [11].

In each independent loop many tube elements are present and each of them has an individual energy balance that contributes to the overall balance mentioned above. Tube dynamics is that of a simple transport delay line with the addition of enthalpy changes due to heat input and pressure changes. Let the transit time t_r be the time that a fluid particle that is at the outlet at time t has spent in the interior of the tube section. In general, if the mass flow rate varies, the transit time will be a function of time, $t_r = t_r(t)$. For each tube with a spatially uniform mass flow rate $\dot{m}_{\text{tube}}$ and constant helium volume V , the transit time can be calculated from the equation

$$V = \int_{t-t_r(t)}^t \frac{\dot{m}_{\text{tube}}(t')}{\rho(t')} dt'. \quad (4)$$

The energy balance equation of a fluid particle that at time $t - t_r(t)$ was at the inlet of the tube and at time t is at the outlet allows to calculate the instantaneous outlet enthalpy of the tube $h_{\text{out}}^{\text{tube}}$ at any time t in terms of the inlet enthalpy $h_{\text{in}}^{\text{tube}}$, the heat income and the changes in pressure. If the heat load $\dot{Q}_{\text{tube}}(t)$ (in W) into the tube, containing at time t' a mass $M_{\text{tube}}(t')$ at uniform density $\rho(t')$, is considered uniform all along the tube, it can be shown from basic thermodynamic principles that

$$h_{\text{out}}^{\text{tube}}(t) = h_{\text{in}}^{\text{tube}}(t - t_r) + \int_{t-t_r}^t \frac{\dot{Q}_{\text{tube}}(t')}{M_{\text{tube}}(t')} dt' + \int_{t-t_r}^t \frac{1}{\rho(t')} \frac{dP}{dt'} dt'. \quad (5)$$

Finally, the temperature of the fluid in each tube section, necessary to calculate the convection heat transfer from solid structures, is estimated with the He state relation $T = T(P, u_{\text{tube}})$. The internal energy of a tube element, u_{tube} , is computed using the energy balance equation, where all the dynamic quantities are considered at instant t :

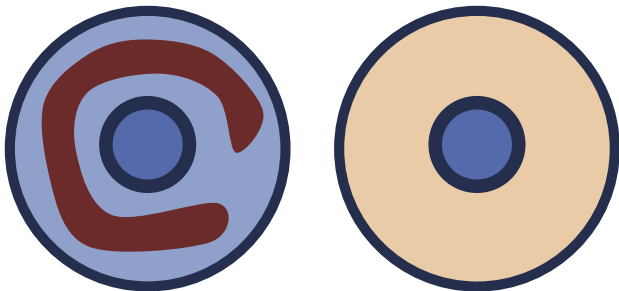


Fig. 3. Types of conductor tube cross-sections. Left, type 'c': the irregular shape represents the solid immersed in the annular channel. Right, type 's': the annular channel does not contain any solid but the heat load that would be exerted on the solid is applied directly on the fluid.

$$\frac{d(M_{tube} u_{tube})}{dt} = \dot{Q}_{tube} + \dot{m}_{in}^{tube} h_{in}^{tube} - \dot{m}_{out}^{tube} h_{out}^{tube}, \quad (6)$$

where $\dot{m}_{in}^{tube}$ and $\dot{m}_{out}^{tube}$ are the inlet and outlet mass flow rates for that particular tube (equal after taking account of the constant uniform density modeling assumption).

2.2.3. Connection nodes

Connection nodes act as ideal mixers. When the volume of a node is null, the outlet enthalpy is calculated as the instantaneous average of the inlet enthalpies:

$$h_{out}^{node}(t) = \frac{\sum_i \dot{m}_i^{node}(t) h_i^{node}(t)}{\sum_i \dot{m}_i^{node}(t)}. \quad (7)$$

If the volume of the node is different from zero, the element acts as a volume mixer, where the incoming fluid gets perfectly mixed instantaneously with the bulk fluid in the element and the outlet enthalpy is that of the bulk:

$$\frac{dH_{node}}{dt} = \sum_i \dot{m}_i^{node}(t) h_i^{node}(t) - h_{node}(t) \sum_o \dot{m}_o^{node}(t) + \frac{dP}{dt} V_{node}, \quad (8)$$

$$h_{out}^{node}(t) = h_{node}(t) = \frac{H_{node}(t)}{M_{node}(t)}, \quad (9)$$

where H_{node} is the total enthalpy inside the volume of the node and M_{node} the mass, the sum with index i represents the total rate of incoming enthalpy, the sum with index o represents the total outgoing mass flow rate, P is the pressure of the system and V_{node} is the volume of the mixer. In terms of spatial description, a node as the one described by Eqs. (8) and (9) is 0-D. The time transfer function from any of the inlet enthalpies to the outlet enthalpy is that of a first order delay.

2.3. Numerical scheme

In order to obtain the new values of temperature, energy, or other quantities for time step n , an explicit method has been implemented. Time stepping is adjustable so that it can be increased or decreased based on the system transient. This feature dramatically increases the computational efficiency of the model.

Given that the thermal interaction between parts in a cross section relates the differential amount of energy to add to an element with the temperatures at t_{n-1} of other elements, the calculation is done in two steps:

1. Step 1
 - a. Calculation and storage of energy exchanged between parts using temperatures in t_{n-1} .
 - b. Calculation of the overall power balance of each part including heat loads.
2. Step 2
 - a. Update of element temperature using the energy balance.
 - b. (For tubes and tube connection elements only) Calculation of the new value of the outlet enthalpy.

For calculating the energy provided by heat loads during the n -th time step, the heat load function is evaluated at $t = \frac{t_{n-1} + t_n}{2}$. Consequently, Eqs. (1), (6), and (8) become discrete differential equations for time advancement of the different types of elements (with the constant density model):

Solid number i :

$$M_i^{solid} C_i \frac{T_{i,n} - T_{i,n-1}}{\Delta t} = - \sum_j S_{ij} (T_{i,n-1} - T_{j,n-1}) - \sum_k E_{ik} (T_{i,n-1} - T_{k,n-1}) + F_i \left(\frac{t_{n-1} + t_n}{2} \right), \quad (10)$$

Tube number k :

$$M_k^{tube} \frac{u_{k,n} - u_{k,n-1}}{\Delta t} = E_{ik} (T_{i,n-1} - T_{k,n-1}) + F_k \left(\frac{t_{n-1} + t_n}{2} \right) + \dot{m}_{k,n-1} h_{in,k,n-1} - \dot{m}_{k,n-1} h_{out,k,n-1}, \quad (11)$$

Tube connection node:

$$M_{node} \frac{h_{node,n} - h_{node,n-1}}{\Delta t} = \sum_i \dot{m}_{i,n-1}^{node} h_{i,n-1}^{node} - h_{node,n-1} \sum_o \dot{m}_{o,n-1}^{node} + \frac{P_n - P_{n-1}}{\Delta t} V_{node}. \quad (12)$$

For tubes of type 's' or 'c' the right hand side must also include heat exchange with the central channel and eventually the immersed conductor, represented by a term of the same kind: $H_{ij}(T_{i,n-1} - T_{j,n-1})$, where j is this other component and H_{ij} the corresponding heat exchange coefficient.

The numerical scheme was implemented so that independently of how the model is constructed (number and types of tubes, solids, or connectors), the time-stepping can be adjusted to assess the introduction of numerical errors. The model equations and the numerical integration routines were programmed in C language, and the resulting code named STIMS (Simplified Thermo-hydraulic ITER Magnet Simulator). The following sections will expand on the fine-tuning of level of detail needed in STIMS to properly capture the dynamic behavior, and on the code benchmarking based on results of simulated cases.

3. Application of the model and results

The first step in the model development was the definition of all flow and heat transfer features thought to be significant in determining the thermal response of the magnet system (as seen from the cryoplant) to the heat load scenarios coming from plasma operations. The derived, most complete, model was described in Section 2. The next step was to quantify the effect of including, or not, each of the flow and heat transfer features on overall magnet system thermal dynamics. This is best done on a coil that has all the features, such as flow pipes of different types, fluid–solid interactions, counterflows, etc. Therefore, determination of the effect of each model feature was done by simulating a plasma scenario on the TF coils. This first effort on model fine-tuning is presented next, in Section 3.1. The next and final step was to apply the simplest model, retaining only the most important features as explained in Section 3.1, to the simulation of thermal transients in the CS coil. This is presented in Section 3.2. The order of presentation of these two sub-sections then follows the progression used in model development and fine-tuning.

3.1. Analysis of model features using ITER TF coils

The results presented in this section show how the model was fine-tuned by sequentially adding modeling features in order to quantify the effect of model complexity on the accuracy of the results. The ITER Toroidal Field (TF) coils, subjected to nuclear heating during plasma operations, were used as an example to assess the impact of each feature of the model on the accuracy of the results and on the code performance.

3.1.1. Modeling of TF coils

The TF coil assembly is a D-shaped coil with a straight leg and a curved leg. The TF magnet is composed of the Winding Pack (WP) and the Casing. A more detailed description of this system can be found in the work by Mitchell et al. [7].

The WP is formed by 7 double pancakes per coil that contain stainless steel radial plates (RP) and radial covers (RC) in between

the CICC's. The two circuits in one double pancake have alternate flow directions, called clockwise (CW) and counterclockwise (CCW), and the number of turns in each pancake depends on its position in the cross section (11 in general, 9 or 3 at the sides). Furthermore, the TF coil cross section is different in the inboard straight leg or in the outboard curved leg. Fig. 4 shows the actual TF cross section of the straight leg. The WP is surrounded by the casing structures. Higher heat loads are expected on the casing which faces the plasma. The casing is cooled by a number of tubes disposed on its surface in contact with the winding pack.

To build a simplified model of the TF coils, the magnet is divided into a certain number of *magnet elements* in the direction along the flow so as to represent with some level of detail the heat load distribution and reinforce the validity of the hypothesis of uniform temperature in each of these zones. Fig. 5 illustrates this discretization. Arrows at 6 and 12 o'clock show the inlet (bottom) and the outlet (top) of the casing cooling circuit. Thus, the flow direction is upwards in both curved and straight legs of the magnet. The arrow in the 32nd element shows the point at which are situated the inlet feeder and outlet collector of the WP circuits. For both CW and CCW loops, they start and end both at this bottom location.

With the objective of keeping a certain degree of complexity to represent the main features of the heat distribution along the path of the coolant, but using a minimum number of elements, a series of simplifications were made:

- The cooling of the WP has been reduced to only two pancakes, one CW and the other CCW. Each one transports the total mass flow of the 7 pancakes it represents.
- The fact that the number of turns for border pancakes is lower (3 and 9 turns) was ignored. The total helium volume in the coil and the total mass flow rate, including those of the shorter pancakes, were divided in 2 equal parts to fit the CW-CCW description, without losing volume or mass flow rate.
- The casing is divided into only three solids: Plasma facing, side and external structures.
- The several cooling conducts of the casing were reduced to only one equivalent tube element in each of the three casing zones.

The schematic of the resulting cross section after these simplifications is shown in Fig. 6. Internally, the different parts interact thermally between each other if they are in contact. Thermal contact coefficients have been estimated using a thermal resistance approach and taking into account the combination of several

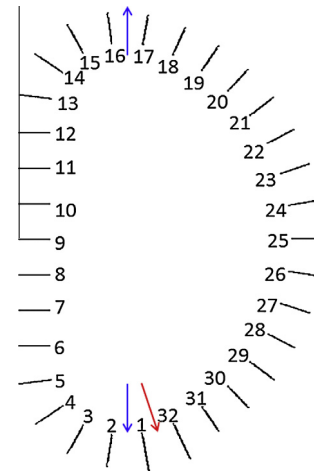


Fig. 5. Discretization of the TF coil.

elements into a single one (by using equivalent overall heat transfer coefficients).

Within this scheme each of the TF *magnet elements* can be regarded as a black box where the incoming signals are the inlet flow properties (mass flow rate and enthalpy) in each of the channels and the heat loads on every element of the cross section. The output from the black box is the outlet enthalpies. The schematic of this black box is represented in Fig. 7.

Furthermore, the magnet elements were hydraulically connected in series. To visualize the approach, let us assume that the TF Assembly magnet has been divided into only three elements and that the CW and CCW loops have only four turns. For the CW loop, and given that after going through the three elements, the coolant flowing through the conductor enters the next turn of the spiral winding, the outlet of the first turn in the last element has to be connected to the inlet of the second turn of the first element. For the CCW loop, the outlet of the first turn in the first element is the inlet for the second turn in the last element (because the flow direction in these tubes is opposed to the CW ones). This simplified example is shown in Fig. 8, where the casing cooling circuits have not been represented for simplification. In the model with 32 elements, casing circuits are connected in ascending order for the elements between 2 and 16 and descending order for the elements between 17 and 32, and 1. Global inlets are in elements 2 and 1, and global outlets in elements 16 and 17. So in fact there

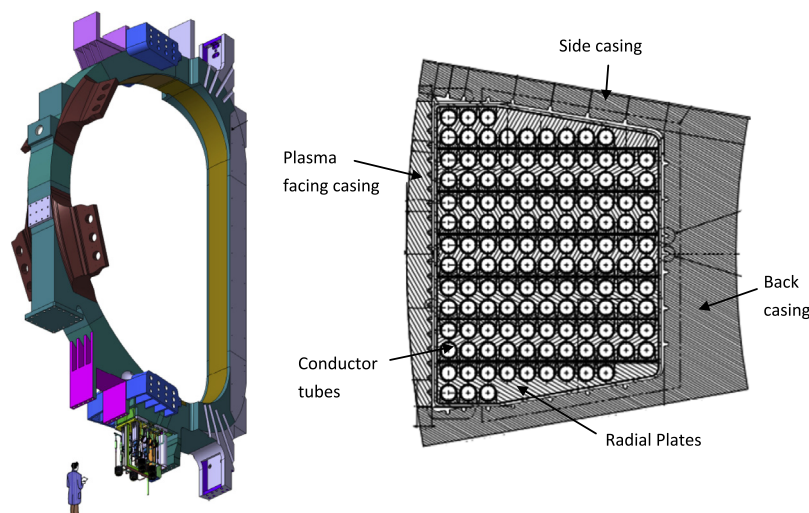


Fig. 4. TF coil. Left: TF coil assembly with all structures in contact. Right: cross section of the TF coil along the straight inboard leg.

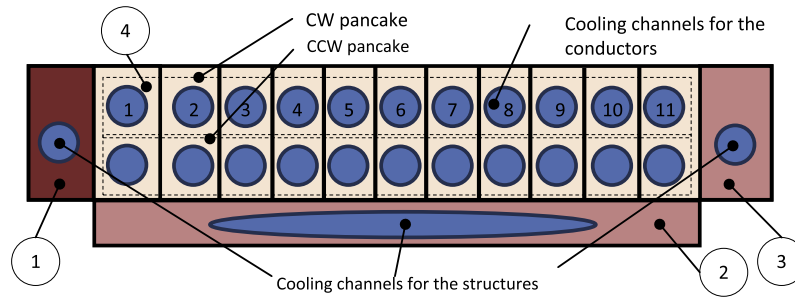


Fig. 6. Cross section of the simplified model for TF coils. References: 1. Casing plasma facing structure; 2. Casing side structure; 3. Casing external structure; 4. Radial plates.

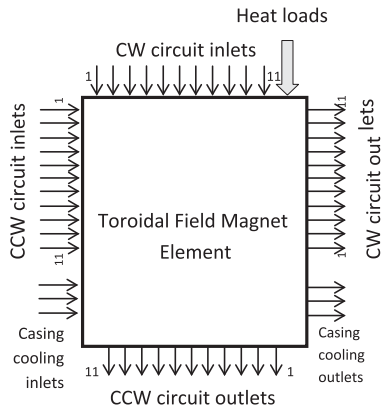


Fig. 7. Black box representing a magnet element.

are 6 circuits in the casing, three on the straight leg of the D and three on the curved leg.

With this somewhat complex network the model captures all the correct interactions between hydraulic channels and thermal masses in the TF coils, so that a purely mass transit approach is still accurate.

3.1.2. Scenario description

The main heat load applied to the TF WP is nuclear radiation stemming from plasma burn. This radiative heat load is active only during 400 s every 1800 s of operation (corresponding to a plasma pulse burn cycle). During this period the value of the power can be considered constant in time. The spatial distribution, according to

numerical calculations that have been performed to assess the magnitude of nuclear radiation incidence on the ITER TF coils, has a very intense peak at the centre of the straight leg of the magnet. Figs. 9 and 10 depict the nature of time and space distribution of the nuclear heat loads (NHL), which can be approximated by the expression:

$$\text{NHL} = G + A e^{-\frac{(x-x_m)^2}{2\sigma^2}} \quad (13)$$

where G is a base value of the heat load, A is the amplitude of a Gaussian bell, x is the element index in the 32-element magnet discretisation scheme along the flow direction (see Fig. 5), x_m is the index of the element where the peak is centered and σ is the width of the peak using as units the segments of discretisation. The values of G and A are defined for each winding turn individually, by fitting the calculated data to Eq. (13).

A scenario where only the nuclear heat load is active will be used to assess the influence of different modeling features and parameters. The shape of the time response to this heat load will be a convolution of the transit phenomena with the heat exchange between the fluid and the surrounding solid structures.

The transit time of the fluid along one turn of coil is about 220 s (as it can be calculated using volumes and mass flow rates). Thus, all the particles of fluid will pass by the NHL peak zone during the duration of the NHL pulse and most of them will also pass by the peak in the following coil turn. However, a small fraction of them, depending of their position in the circuit when the NHL pulse starts, will only pass through a fraction of the peak in the next turn or could even not reach this second peak at all (because $2 \times 220 \text{ s} > 400 \text{ s}$, see Fig. 10 for better understanding). This, combined with the fact that different turns have different heat load

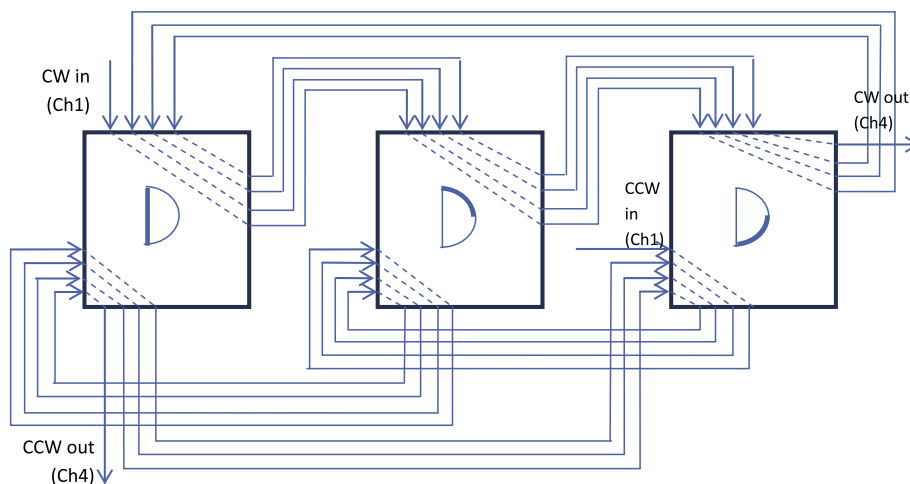


Fig. 8. Example of connection between the cooling conducts of different magnet elements.

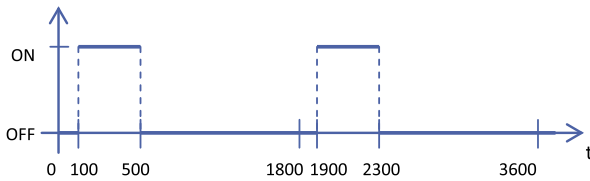


Fig. 9. Nuclear heat load status along time. 400 s long pulses every 1800 s.

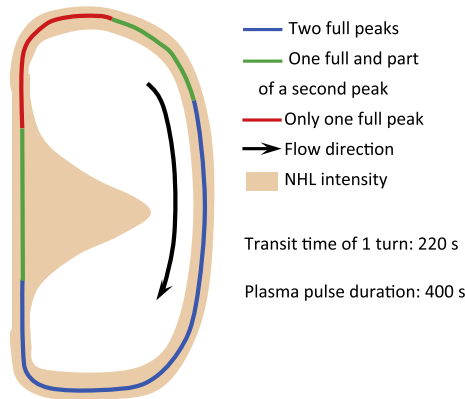


Fig. 10. Number of nuclear heat load (NHL) peaks a helium particle will transit through during NHL-ON interval according to its position at the moment when NHL turns ON.

intensity, will define the time response of the enthalpy and temperature at the outlet of the circuit.

3.1.3. Pure transit

The simplest model to consider is one whose tubes are type 'z' and without heat exchange between the fluid and the radial plates, i.e. adiabatic tubes with a bulk heat source, the NHL. As heat diffusion along the direction of the flow is absent in the presented model, the time evolution of the outlet properties is the direct result of the accumulation of energy, along the whole WP, by the fluid particles present at the outlet at each instant. Particles that have received heat load in the inner most turns will be hotter than those that were at the outermost turn when the heat load was present. And since the transit time from innermost turns to the outlet is longer, hotter particles arrive later. This behavior is exactly what is observed in Fig. 11A, which shows a plot of the outlet temperature of the CW coils in a simulation of NHL on WP with adiabatic (no heat exchange with radial plates) 'z' tubes and in open loop (OL) configuration.

Given the spiral configuration of the TF hydraulic circuit one pulse of heat load concentrated in time and space produces a succession of plateaux and valleys in the outlet temperature distributed over a longer time scale. Flat plateaux correspond to those particles that have passed by two entire NHL peaks during irradiation and small valleys separating the plateaux correspond to that little fraction of fluid that only goes through one heat load peak during the irradiation. The soft borders of plateaux are linked to those particles of fluid that encounter a varying fraction of a second NHL peak during irradiation. The plateau around $t = 2600$ s, which is lower than the one before it, corresponds to those particles that were heated only in the innermost turn of the pancake, because at the beginning of the irradiation they were not yet in the magnet and did not reach the NHL peak in the second turn during irradiation. It should be noted that the highest outlet plateau ends about 2420 s after the start of the heat load pulse (at $t = 100$ s), and this is the transit time along the pancake set up on the simulation (11 turns, 220 s/turn). Everything indicates that the transit phenomena

are well represented here. As pulses are repeated every 1800 s, less than transit time along a pancake, a fraction of the fluid, which is at the exit between $t = 1900$ s and $t = 2300$ s, has gone through a second irradiation and thus has a minor heat contribution from the last turns of the pancake during the second heat load pulse.

3.1.4. Effect of heat exchange with radial plates

The model is then used to analyse the mixed outlet of the CW and CCW circuits in the presence of heat exchange between the fluid in the tubes and the radial plates (Fig. 11B). A parametric analysis is performed by changing the value of this parameter from 0 to $10 \text{ W/m}^2 \text{ K}$. A smoothing of the response to a NHL pulse is visible, due to heat exchange with the radial plates before the fluid reaches the outlet. For low values of the heat transfer coefficient, the temperature steps can still be observed, but the amplitude of the peaks in the overall response starts to diminish since there is now another element storing energy (the radial plates) that takes part of the heat from the hot fluid and has a certain inertia to return it to the cold fluid. For a sufficiently high coefficient, the steps in the response are no longer observed. Furthermore, it is seen that thermal coupling with the radial plates adds to the dynamics a slow transient until reaching the periodic steady state, and the time constant of this transient, lasting several pulses, is related to thermal inertia of the radial plates.

3.1.5. Pressure evolution

For all closed loop (CL) simulations pressure evolution is modeled through Eqs. (2) and (3). The temperature response is similar but there is an extra contribution to it due to changes in pressure (presented in Fig. 11C for all the cases discussed in the previous sections). During the 400 s corresponding to each NHL pulse, for the CL curve in Fig. 11A the plateaux in the outlet temperature evolution show a positive slope, due to the fact that energy is being absorbed by the fluid and the system is pressurizing everywhere (pressurization causes a rise in temperature at operating conditions). It is also during this period of time that OL and CL results separate visibly. During the NHL step, energy balance and pressure's derivative in time are positive. But when NHL is not present, the absolute value of the pressure decrease rate is well correlated to the temperature at the entrance of the heat exchanger (average outlet temperature of the CW and CCW circuits), since the hotter the fluid, the higher the driving temperature difference in the heat exchanger. This suggests that the heat load for the cryoplant is approximately proportional to the difference between the entrance temperature and the heat exchanger cold side temperature.

The heat exchange with the radial plates limits the amplitude of temperature variations and, in accordance to last paragraph, it limits the amplitude of pressure oscillations too, as depicted by the non-solid curves in Fig. 11C.

3.1.6. Effect of the conductor tube details

The same scenarios presented in the previous sections were analyzed with a model based on conductor tubes of type 'c', i.e. including a detailed tube description with a solid for conductor and a central channel. The analysis was made in two separate and distinct steps:

- Studying the effect of the presence of the conductor, qualitatively and quantitatively.
- Studying the effect of the presence of the central channel.

3.1.6.1. Effect of the presence of a conductor. In order to observe only the effect of the presence of conductor, volume and mass flow rate have been distributed between the two channels in equal fractions, so that both channels have the same fluid velocity and transit time.

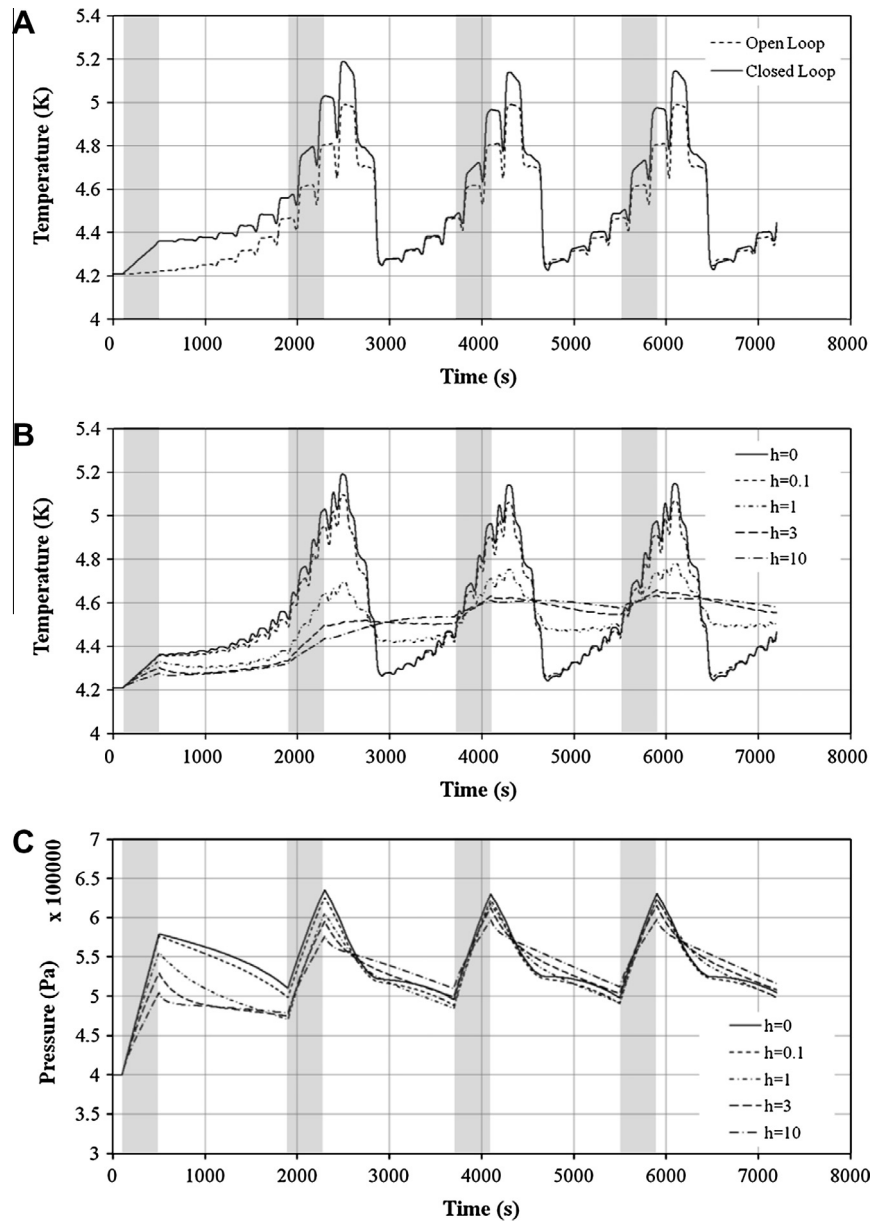


Fig. 11. (A) Outlet temperature of fluid in the CW winding with no heat transfer. (B) Outlet temperature of WP collector with heat transfer to the radial plates (h in $\text{W/m}^2 \text{K}$) in closed loop. (C) Pressure evolution in the TF WP when the loop is closed for different values of heat transfer coefficient with RP (h in $\text{W/m}^2 \text{K}$). In the three subfigures the grey areas represent the instants of time where the nuclear heat load is present.

Thus they do not represent ITER magnets accurately, where the annular channel volume is 81% of the conductor helium volume whereas the mass flow rate is distributed in equal parts between the two conducts (the velocity in the central channel is around 4 times higher than that in the annular channel for the ITER TF coils).

The two parameters that play a role in the conductor effect are its heat capacity and the heat transfer coefficient between the fluid in the cable annulus and the conductor.

The mixed temperature at the outlet of the CW and CCW loops was evaluated using three versions of the model in open loop and with no heat transfer to radial plates:

- Model without conductor (previous section results).
- Model with conductor having the nominal heat capacity for ITER TF coils.
- Model with conductor having around 25 and 50 times higher heat capacity.

For this study, the value of the heat transfer coefficient between the conductor and the fluid has been set to a value of $C = 100,000 \text{ W/K}$, as it is defined in the program. The values of this coefficient are calculated as the surface-integrated heat transfer coefficient, i.e.,

$$C = hPLN \quad (14)$$

where C is the coefficient, h is the heat transfer coefficient ($\text{W/m}^2 \text{K}$), P is the wetted perimeter of the conductor, L is the length of a winding turn of only one tube and N is the number of TF coils, 18. It should be noted that a value of $100,000 \text{ W/K}$ for C corresponds to a value of $\sim 70 \text{ W/m}^2 \text{K}$ for h . The hydraulic diameter Reynolds number in the annular channel is around 2300. Consequently, an almost laminar value of Nusselt number $Nu = 6$ has been taken to evaluate the heat transfer coefficient. Thus, the ITER TF coils have an estimated value of $C = 400,000 \text{ W/K}$ or $h = 280 \text{ W/m}^2 \text{K}$, as can

be deduced from present ITER coils design values of $P = 2.25$ m and $L = 35$ m. These values of wetted perimeter and turn hydraulic length can be estimated from the data in the articles [2] and [5].

The results of these simulations are shown in Fig. 12. Comparing the two curves corresponding to the cases with exaggerated conductor heat capacity, it is noted that the presence of a solid immersed in the tube introduces a time delay in the arrival of energy to the outlet of the magnet cooling circuit. As the fluid enters cold in the tube, it starts exchanging heat with a hot solid (innermost turns, high heat load). This now hot fluid, as it advances in the loop, reaches a region where the solid has not been intensely heated (the outermost turns) and is colder than the fluid, which therefore drops its temperature, while the solid heats up. Once the conductor in this region has been sufficiently heated, its temperature is closer to that of the arriving fluid and the heat exchange between the solid and the fluid starts to decrease. A certain time is required by the solid near the exit to reach a temperature similar to the fluid temperature and stop cooling down the arriving fluid. This is a thermal inertia effect and the time delay constant will be linked to the solid thermal capacity. This is seen in Fig. 12, as the conductor mass increases, the time at which the peak occurs is delayed by an amount of time that is in a linear relation with the heat capacity. It is also observed that as the time delay increases there is a decrease in the temperature peak.

However for actual ITER coils, which operate at a temperature close to 4 or 5 K, the volumetric specific heat of copper and superconducting materials is much lower than that of helium. Thus, as it is seen in the figure, the effect of including the conductor in the

model is not significant. It is necessary to mention that including it has an important computational cost because it forces the simulation to adopt a very short time step value in order to keep the explicit numerical scheme stable. Whenever this feature of the model does not really introduce new information it should be suppressed.

The effect of the convection heat transfer coefficient between the fluid in the annular channel and the conductor was studied by means of a parametric analysis. Different orders of magnitude of this coefficient were set in the model and the corresponding response of the outlet temperature is shown in Fig. 13.

This coefficient, at sufficiently high values, does not have an effect on the delay time of the power pulses. Its effect consists mainly of allowing the propagation of heat load spatial irregularities to the outlet temperature. When the coefficient is large enough (which is generally true given the results observed in Fig. 13), the filtering of the heat load signal to the heat transferred to the fluid is negligible. When the coefficient is low enough, the heat transfer from the conductor to the fluid has a time delay that is larger than the transit time associated with the size of these irregularities, and so they are filtered away. However, as it can be observed, in practical cases of superconducting magnets, the exact value of this parameter is irrelevant as long as it is high enough. Thus, it is neither necessary nor influential for codes simulating this kind of systems to devote much calculation time to an accurate determination of this parameter by means of correlations for thermohydraulics purposes.

This analysis shows that the inclusion of the conductor in the tube is necessary only if the thermal inertia of the conductor

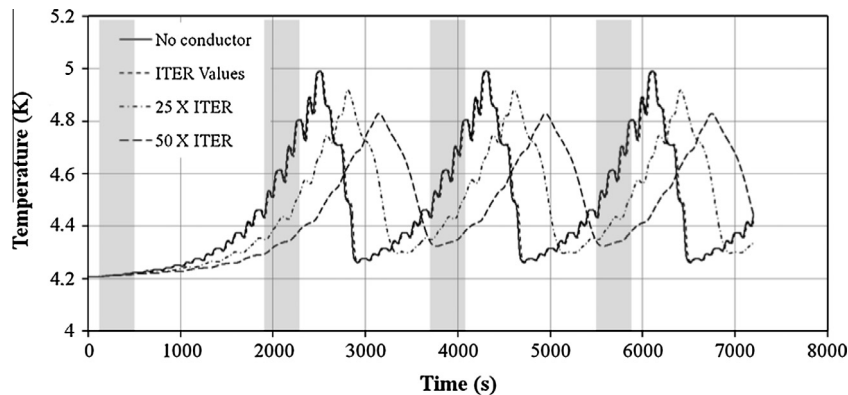


Fig. 12. Parametric analysis of conductor heat capacity in tubes of type 'c' with equal velocity in annular and central channels. The curves represent the temperature at the exit of the collector of the WP's. The grey areas represent the instants of time where the nuclear heat load is present.

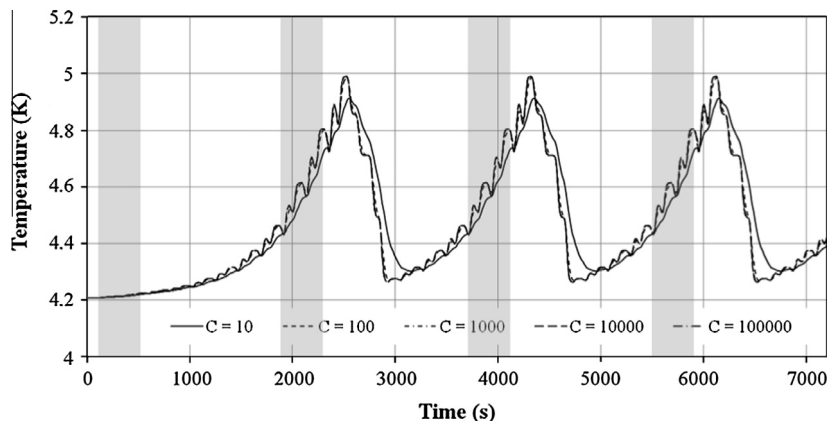


Fig. 13. Parametric analysis of conductor heat transfer coefficient in tubes of type 'c' with uniform velocity. Curves with C (in W/K) higher than 100 are all superimposed. The curves represent the temperature at the exit of the collector of the WP's. The grey areas represent the instants of time where the nuclear heat load is present.

material is considerable compared to the helium heat capacity (not the case in ITER). The effect of the heat transfer coefficient, at such high values as ITER's, is not influential on the outlet temperatures or enthalpies.

3.1.6.2. Effect of the central channel. In order to assess the effect of the presence of a central channel, two simulations were carried out. The first considers the presence of a central channel and the volume fractions for annular and central channels are fixed at the same values as mass flow rate fractions, thus having uniform fluid velocity in the two channels. The second simulation was performed with a value of 19% for the volume fraction of the central channel, which is the approximate value in the ITER TF conductor, while the mass flow rate is divided into two equal parts, which is also approximately true for ITER TF coils. The resulting temperature evolutions (at the exit of the winding pack collector) are plotted in Fig. 14.

When the velocity of the central channel is higher than in the annular channel, there is a homogenization of the temperature along the flow direction due to the breaking of spatial synchronisation of irregularities. The heat transfer coefficient between the two channels is high and the transient of temperature equalization is

very fast. When the fluid speed is uniform, the conductors heat all the fluid in both channels simultaneously, as their temperatures at each point of the circuit are the same. Irregularities in the two channels are synchronized and once there is no more heat source, no more heat transfer occurs between the two channels. But in the second case, when the irregularity fronts in the central channel travel faster, the heat exchange between the concentric channel smoothes them away. However, if irregularity patterns are not so pronounced as in this case (but rather by coil turn, as in the case of the ITER Central Solenoid coils, presented in next section), the inclusion of the central channel in the model could be obviated, as its effect is imperceptible.

3.2. Benchmarking using the CS coil

The Central Solenoid (CS) of ITER is responsible for creating an inductive toroidal electric field to drive a current in the plasma. This is achieved by varying the magnetic flux going through the CS. The CS coil is formed by six identical stacked modules. This magnet system presents revolution symmetry. A schematic of the CS system is shown in Fig. 15. The uppermost and lowermost modules are in thermal contact with structures which do not directly

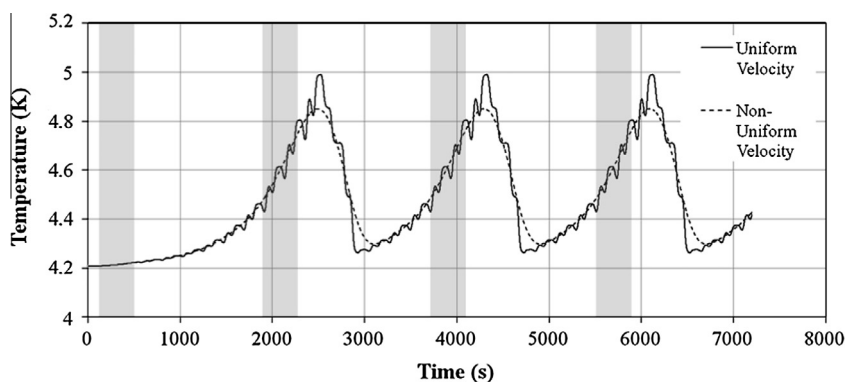


Fig. 14. Effect of the central channel of conductor tubes when velocity is higher than in the annular channel. The curves represent the temperature at the exit of collector of the WP's.

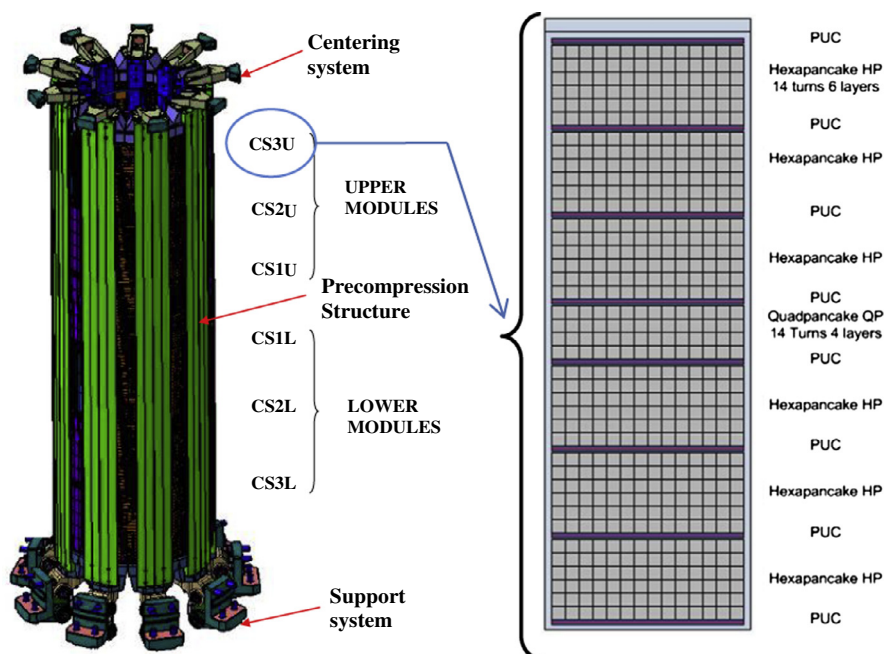


Fig. 15. Drawing of the Central Solenoid and scheme of multi-pancakes in each module.

constitute a part of the winding pack and this could be relevant to the modeling.

The CS coil has already been modeled in Vincenta [12] and SuperMagnet [13], two standard and well-established codes used for the thermo-hydraulic simulation and conductor margin estimation of ITER magnet operating scenarios. Since the results of these simulations are available, and the codes have been benchmarked, it is possible to use these previous results to assess the accuracy and performance of the current model.

3.2.1. CS coil model

Each CS module is a set of six hexa-pancakes and one quadri-pancake (this denomination refers to the number of pancakes that are electrically in series, 6 or 4 respectively). Thus, each module has 40 pancakes. Hydraulically, as for the TF coils, alternate pancakes have alternate flow directions (CW and CCW). Fluid inlet is always at the innermost turn of each pancake. Each pancake has 14 turns of a CICC with round-in-square steel jacket (Fig. 2). Turns are separated from each other by a 3.5 mm insulation layer, whereas hexa and quadri-pancakes are insulated from each other by a thicker layer. Finally modules can be considered thermally isolated from each other.

3.2.1.1. *CS heat loads.* The heat loads on the CS are, per pulse:

- 8115 kJ for the AC loss contribution.
- 2572 kJ for the feeder and cryoline contribution.
- 66.3 kJ for the Ohmic heating in the joints.
- 1.8 kJ for the AC losses in the joints.

Only the first two sources are included in the modeling, while the others are neglected.

Cryoline heat loads are considered uniform in space along each tube and constant over time, but AC losses are more complex and vary by pancake, turn number, and time. The time dependence is represented by step functions based on the variation of the current in the CS, with a substantial peak at the beginning of the pulse and concentrated in the innermost turns.

3.2.1.2. *Simplified CS cross section.* For the sake of simplification, the contact between CS and the top and bottom structures is assumed

adiabatic, as has been done before for the Vincenta and SuperMagnet models.

Fig. 16 shows the simplified CS cross section composed of six stacked modules. The 40 pancakes of a single module are condensed into just two equivalent pancakes, CW and CCW, each containing 14 turns and with the inlet at the innermost turn. The 14 turns were kept as separate elements each with a different heat load because the heat load distribution is strongly inhomogeneous in the radial direction. Even if there is insulation between the stacked pancakes, simulations done with other software show that the temperature is quite uniform in the vertical direction, as the heat loads are also uniform. Consequently, heat load time-space profiles have been simplified accordingly to two alternate flow direction pancakes per module and there is only one solid interacting with the two counter flow pancakes.

3.2.1.3. *Hydraulic network.* The CS hydraulic model consists of six thermally independent double pancakes interconnected on the same hydraulic network by means of the feeders and cryolines. The hydraulic network external to the magnet has been represented as close as possible to the one developed in previous Vincenta simulations for CS and is shown in Fig. 17. This way of treating the hydraulic network is very similar to what has been done for calculations of other ITER magnets [1,3].

The six supply and return feeders are all subjected to the same heat load. However, the upper cryolines have different size and heat loads with respect to those of the bottom cryolines. Mass flow rate has been assumed constant in time and identical for the 6 modules both for the CW and CCW pancakes.

3.2.2. Comparison with Vincenta and SuperMagnet codes

In order to perform a complete benchmarking process, different models are considered, with different degrees of simplification to determine the minimum set of elements to include without losing the essence of the magnet system thermal dynamic response. The CS model described above was further simplified, and instead of including six modules, only one is considered, with CW and CCW pancakes. Two different versions of this one-module model were implemented, one using 's' tubes, and one 'z' tubes. In summary, this section presents results from five different models with different levels of detail:

- STIMS Mod6: CS with 6 modules and tubes type 's'.
- STIMS Mod1: CS with 1 condensed module and tubes type 's'.
- STIMS Mod1 "s": CS with 1 condensed module and simplified tubes type 'z'.
- Vincenta: Model with 240 pancakes.
- SuperMagnet: Model with 240 pancakes.

The main differences between STIMS and the other two more complex models are:

- In STIMS, the mass flow rate is constant in time at the nominal design value whereas the other two models include dynamics for a pump driving flow in the circuit and a friction loss model (two elements completely absent in the presented model). However, during most of the transient simulated in Vincenta or SuperMagnet the mass flow rate is close to its nominal value.
- The pump in Vincenta and SuperMagnet introduces energy before the fluid enters the heat exchanger.
- There is a by-pass valve across the heat exchanger in the Vincenta and SuperMagnet models.
- In Vincenta and SuperMagnet, helium is compressible.

Figs. 18–21 show a summary of the results of flow conditions at the magnet and cryoplant inlets (or at the point equivalent to the

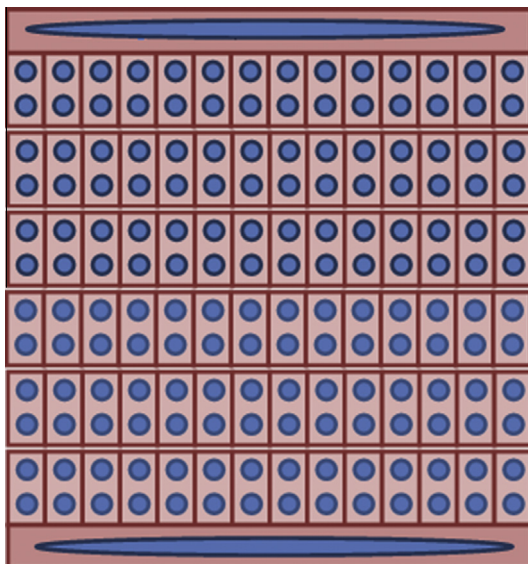


Fig. 16. Cross section of the simplified CS model.

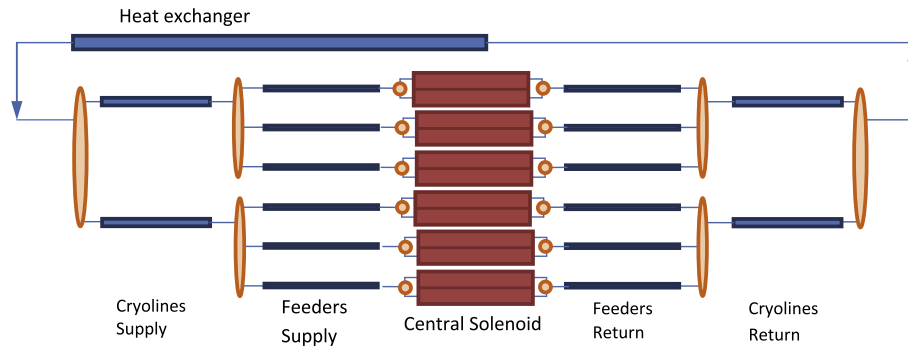


Fig. 17. Complete hydraulic network as represented in the CS model.

outlet of the last collector, at the very right of Fig. 17) simulated with Vincenta, SuperMagnet, and STIMS for the first four plasma pulses starting from a uniform temperature condition.

The first observation is that the three STIMS models whilst characterized by a different degree of simplification exhibit exactly the same behavior to the point that their curves are not distinguishable. This is a very encouraging result, since it shows the possibility of drastically simplifying the model without compromising the accuracy of the results of interest at the magnet/cryoplant interface. Simulations with tubes of type 'c' have been performed too, though not shown in the figures, and the results are also indistinguishable from those presented here.

Secondly, the pressure calculated by STIMS is not evaluated at the outlet of the system, but it is the volume-averaged pressure of the circuit. This is the reason why *absolute* pressure values are

not necessarily equal in all three models. In fact, the pressure at the entrance of the cryoplant is the lowest of the entire circuit since it is at the inlet of the pump. However, the pressure differential behavior should be equivalent in the models; yet the amplitude of the pressure oscillations predicted by STIMS is smaller than with the other models.

Finally, in terms of temperature and enthalpy evolution, it is seen that STIMS responds to a heat load pulse with a transient very similar to that of SuperMagnet, but generally delayed by about 100 s. This difference could be due to model differences related to compressibility and varying density, or non-constant and non-uniform mass-flow rates. It is necessary to keep in mind that there are various phenomena affecting time history of the CS collector outlet (heat exchanger inlet) enthalpy, and their individual transfer functions are quite different:

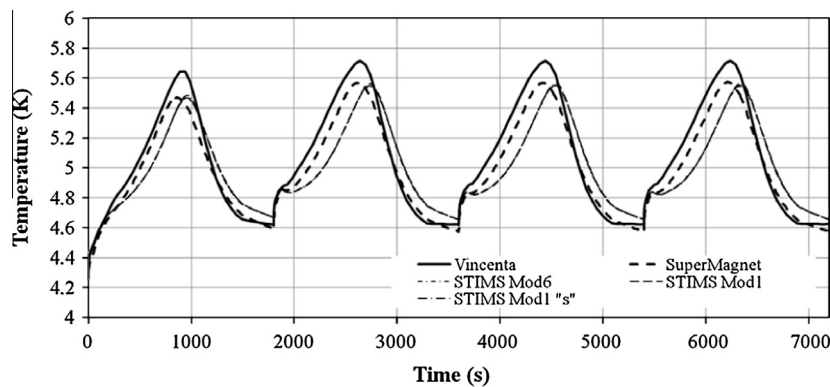


Fig. 18. Temperature at the inlet of cryoplant, all models.

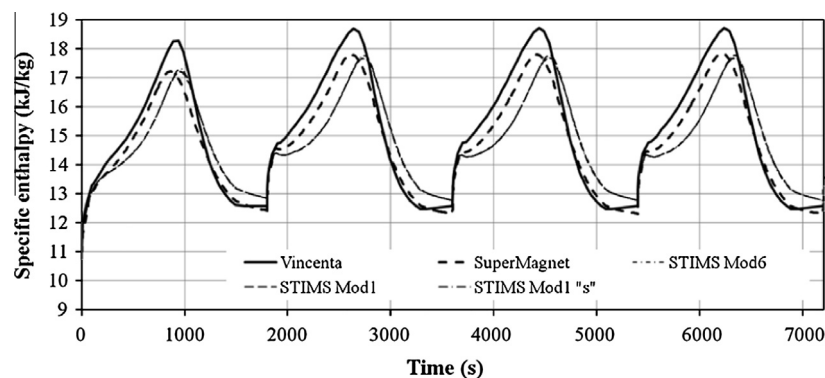


Fig. 19. Enthalpy at the inlet of cryoplant, all models. All STIMS results are superimposed.

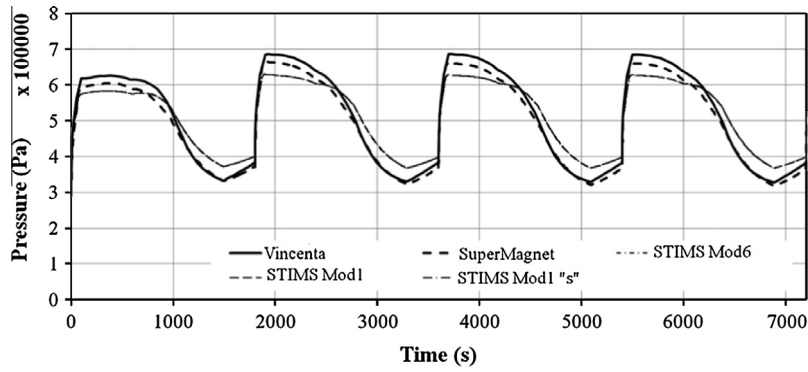


Fig. 20. Pressure evolution, all models. For Vincenta and SuperMagnet this pressure is evaluated at the inlet of the pump, for STIMS it is the average pressure of the circuit.

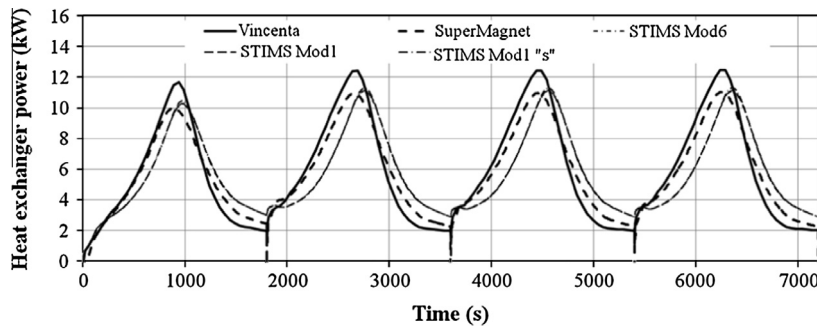


Fig. 21. Power evacuated at the heat exchanger, all models.

- Transit phenomena (pure transport delay).
- Interaction with solids and their thermal inertia (low-band filter delay).
- Pressure effects (instantaneous).

The amplitude of the temperature variations calculated with STIMS is in good agreement with that calculated by the other two models, the largest difference, with Vincenta, being 0.15 K.

The amount of heat evacuated by the cryopant can be calculated as:

$$\dot{Q}_{cryo} = (h_{in}^{cryo} - h_{out}^{cryo})\dot{m} - \frac{dU_{cryo}}{dt} \quad (15)$$

where $\dot{Q}_{cryo}$ is the heat transferred from the cryopant tube to the external helium bath, h_{out}^{cryo} and h_{in}^{cryo} are the outlet and inlet specific enthalpies of the cryopant tube respectively, $\dot{m}$ is the mass flow rate through it, and $\frac{dU_{cryo}}{dt}$ is the rate of change of the cryopant tube total internal energy. The expression above is exact for the incompressible helium model (if not incompressible, there would be different mass flow rate values at the inlet and at the outlet). A further approximation consists of neglecting the last term. As $\frac{dU_{cryo}}{dt}$ has been found to be negligible in almost all the considered

transients, the heat transferred to the helium bath in the cryopant is calculated as:

$$\dot{Q}_{cryo} = (h_{in}^{cryo} - h_{out}^{cryo})\dot{m} \quad (16)$$

Fig. 21 compares the amount of power exchanged in the cryopant calculated by all simulations. The calculation of this quantity using the different models gives similar results, as in the case of the temperature and enthalpy calculation shown before. In order to check the energy conservation, the power was integrated over time for each pulse. The results are shown in Table 1. Naturally during the first and second heat load pulses, the obtained values differ from the integrated magnet and cryolines heat load because during this stage of the transients, energy is being stored in the different circuit elements. But from the third pulse on, the amount of energy evacuated in the cryopant during 1800 consecutive seconds matches the pulse-integrated value of heat loads (about 10.7 MJ). Since this is the most important parameter driving the design of the magnet/cryopant interface (from the cryopant perspective), STIMS provides a very convenient and simplified tool for the simulation of magnet thermal dynamics in response to heat loads due to tokamak operations. A summary of how STIMS simulation results compare with those of Vincenta and SuperMagnet is given

Table 1
Comparison of results from the three models.

	Heat evacuated in HX (MJ)			Peak temperature (K)			Time when the peak occurs (sec)		
	Vincenta	SuperMagnet	STIMS	Vincenta	SuperMagnet	STIMS	Vincenta	SuperMagnet	STIMS
Pulse 1	9.28	9.11	9.463	5.63	5.46	5.465	915	860	958
Pulse 2	10.83	10.66	10.734	5.71	5.56	5.545	2640	2615	2735
Pulse 3	10.88	10.60	10.693	5.71	5.56	5.545	4440	4415	4535
Pulse 4	10.90	10.64	10.692	5.71	5.56	5.545	6240	6225	6335

in Table 1, for the three most relevant parameters: integrated enthalpy at the magnet outlet (total heat load to the cryoplant), as well as peak outlet temperature and time when that peak occurs.

4. Conclusion

A simplified approach for the general problem of simulating flow in internally-cooled magnets has been developed. This approach transforms a magnet system into a hydraulic network of generic cross section structures through the assumption that it is formed by solids with cooling tubes going through them. This model was used to create a calculation tool by means of a C language program, named STIMS. The code allows the simulation of any of the ITER magnets in fairly complicated hydraulic networks, while keeping the details of the metal-fluid thermal interactions at a simplified level. It also allows the user to specify arbitrary heat load scenarios.

A simplified representation for the ITER TF coils with their casing has been implemented in the frame of the simplified model as well as more sophisticated codes. Simulations have been performed in order to assess the significance of the approximations and their impact on simulated behavior. Transit time effects have been evaluated using a nuclear heat load scenario (energy deposition highly concentrated in the straight leg of the TF magnet), and results show how STIMS is capable of properly capture the dynamics of pressure and temperatures in the circuit. The effect of heat transfer between the helium and the inter-tube structures, such as radial plates, has also been evaluated. Moreover, the influence of considering a more detailed description of the tube, with a solid for the conductor and a tube for the central channel has been assessed, and found to be an unnecessary detail for the purposes of this work (specification of temporal variations in cryoplant heat load).

A complete model of the ITER CS coils has been developed and implemented in the code. This model contains the full hydraulic network and a fairly detailed heat load scenario. Results of simulations using this model have been compared with those from Vincenta and SuperMagnet codes. Results show that the simplified model can be effectively used to reproduce the thermal dynamics of the system quite well, obtaining results similar to models that include a largely more complex representation of the hydraulic circuit and conductors/structures. Furthermore, successive simplifications have been applied in order to accelerate the simulations, while ensuring that no critical information on system dynamics is lost in the process.

In summary, the relevant features of ITER magnet thermo-hydraulic modeling have been identified and studied. A black box model has been developed that takes heat loads and helium inlet conditions from the cryoplant and can reproduce, through heat transfer and flow dynamics, the outlet response of the magnet system. The transfer function of the magnet system to multiple input signals as derived by the STIMS model is quite similar to the one that would be obtained with more sophisticated codes through very

costly simulations, unnecessarily detailed for the purpose of this work.

Furthermore, STIMS can accomplish the simulation of a full ITER magnet system over a 2-h period in less than 2 min of CPU on a standard processor, whereas the more sophisticated codes take as much as days to perform the same task. This opens the door to a number of possibilities: rapid conceptual designs related to thermo-hydraulic schemes, magnet/cryoplant interface definitions and simulations, cryoplant control and optimization studies, hardware-in-the-loop testing of cryoplant controllers, and eventually, integration to the actual ITER control logic as a feed-forward element (physics-based predictive control of tokamak operations).

Disclaimer

The views and opinions expressed herein do not necessarily reflect those of the ITER Organization.

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